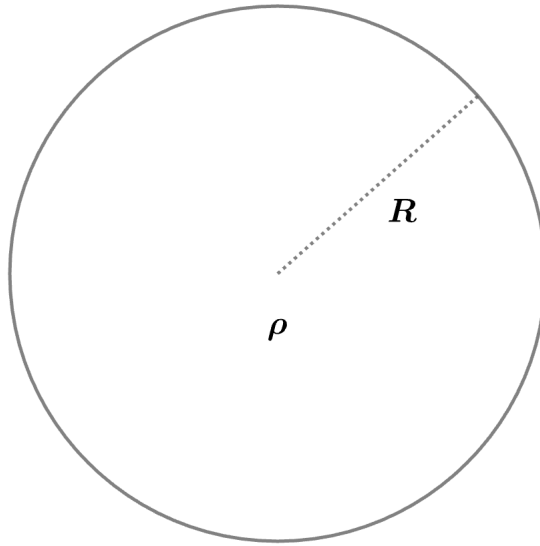


2010 F=ma Exam: Problem 21

Kevin S. Huang



Recall the binding energy of a solid sphere of mass M and radius R is

$$U_{\text{bind}} = \frac{3GM^2}{5R} \propto \frac{GM^2}{R}$$

We could also have obtained the scaling from dimensional analysis. We have

$$M \propto \rho R^3$$

$$U_{\text{bind}} \propto G\rho^2 R^5$$

If the radius is doubled, the energy increases by a factor of $2^5 = 32$ so the answer is E.